FIITJEE

ALL INDIA TEST SERIES

FULL TEST – XIV

JEE (Main)-2021

TEST DATE: 11-07-2021

Time Allotted: 3 Hours **Maximum Marks: 300**

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts.**
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A and Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B :** Do any five questions out of 10 Questions.

  Section-A (01 – 20, 31 – 50, 61 – 80) *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

  Section-B (21 – 30, 51 – 60, 81 – 90) *contains 30 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer. There is no negative marking.*

Physics

PART – A

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. An elastic massless horizontal wire of length 2L is attached to the two fixed supports at the same level. Now a block of mass M is attached at the mid point of the wire. Then the vertical displacement of the mid-point of wire in equilibrium is (Y = Young's modulus of wire, A = cross section area of wire)

(A) $\left[\frac{Mg}{AY}\right]^{1/3}(L)$

(B) $\left[\frac{2Mg}{AY}\right]^{1/3}(L)$

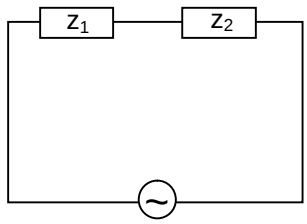
(C) $\left[\frac{Mg}{2AY}\right]^{1/3}(L)$

(D) $\left[\frac{Mg}{AY}\right]^{1/3}(2L)$

2. A projectile is thrown from the ground with a speed of 20 m/s at an angle of 37° with the horizontal, the projectile strikes the ground with a speed of 10 m/s at angle of 53° with the horizontal. The drag force due to air is $\vec{F}_d = -k\vec{v}$, where $\vec{v}$ is the instantaneous velocity. Find the time of flight (Take g = 10 m/s^2)

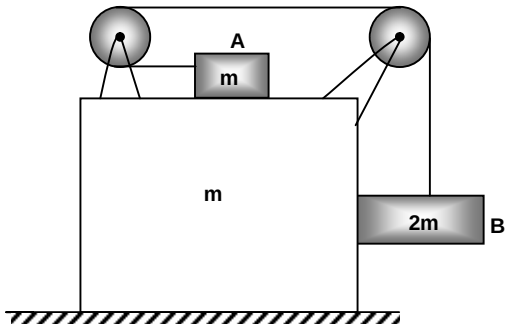
(A) 1.2 sec
(B) 1.6 sec
(C) 2 sec
(D) 2.4 sec

3. Two impedances z_1 and z_2 having their voltage phase difference of 90°. Their resistive component are different and non zero. Choose the correct statement.



(A) Both z_1 and z_2 are inductive in nature
(B) Both z_1 and z_2 are capacitive in nature
(C) One is inductive and other is capacitive
(D) One is resistive and other is either inductive or capacitive

4. The system is released from the rest as shown in the figure. All the surfaces are smooth and strings and pulleys are ideal & massless. The acceleration of block B just after the release is



(A) $\frac{6}{11}g$

(B) $\frac{\sqrt{17}}{11}g$

(C) $\frac{8}{11}g$

(D) $\frac{2\sqrt{17}}{11}g$

5. A glass U-tube has the radius of one limb is r and that of other is 3r. The tube is inverted vertically with open ends below the surface of water (density = ρ) in a beaker. The difference between the heights to which water rises in the two limbs is (surface tension = T and angle of contact between water and glass is 0°)

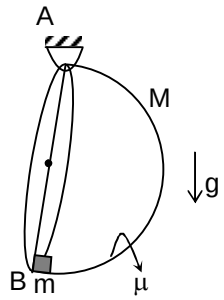
(A) $\frac{4T}{\rho rg}$

(B) $\frac{4T}{3\rho rg}$

(C) $\frac{2T}{\rho rg}$

(D) $\frac{2T}{3\rho rg}$

6. A hollow uniform hemisphere of mass M is hinged from point A in vertical plane is in complete equilibrium along with a particle of mass m that is placed inside the hemisphere at a point B. (which is diametrically opposite to the point A) and The inner surface of the hemisphere is rough. If the particle is just about to slide at this instant. Find the coefficient of friction of the inner surface of the hemisphere.



(A) $\frac{M}{(m+2M)}$

(B) $\frac{M}{2(m+2M)}$

(C) $\frac{M}{(2m+M)}$

(D) $\frac{M}{2(2m+M)}$

7. A positive charged particle is given a velocity $\vec{v} = 3v_0\hat{i} + 4v_0\hat{j}$ from the origin. There is a uniform electric field $-E_0\hat{i}$ and uniform magnetic field $B\hat{i}$ is present in the space (gravity free). Find the minimum value of magnetic field so that particle again passes through the origin.

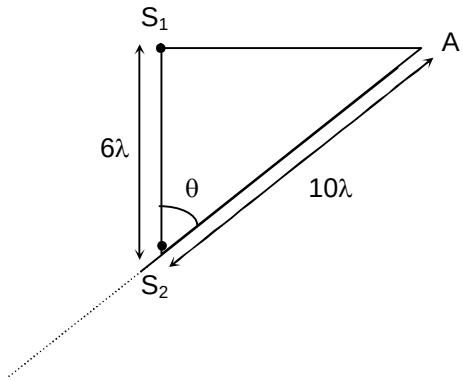
(A) $\frac{\pi E_0}{4v_0}$

(B) $\frac{\pi E_0}{3v_0}$

(C) $\frac{3\pi E_0}{4v_0}$

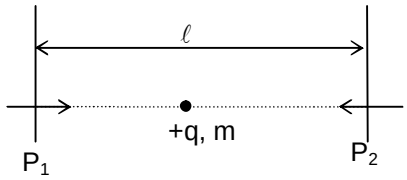
(D) $\frac{2\pi E_0}{3v_0}$

8. A screen S is placed at an angle of $\theta = \cos^{-1}\left(\frac{3}{5}\right)$ with the line joining two coherent sources (point sources) of light of wavelength λ. Find the number of minima observed on the line (from A to infinity as shown) on the screen.



(A) 9
(B) 10
(C) 11
(D) 12

9. Two electric dipoles having dipole moments P_1 and P_2 are fixed at a separation of ℓ such that $\vec{P}_1$ and $\vec{P}_2$ are anti-parallel to each other as shown in the figure. A positive charge particle is slightly displaced from its equilibrium position along the line joining them and released, then time period of small oscillations of the charged particle does not satisfy.

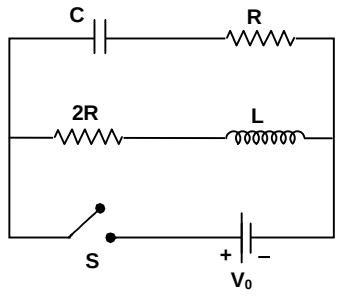


(A) $T \propto \ell^2$
(B) $T^2 \propto \frac{\varepsilon_0 m}{q}$
(C) $T \propto P_1^{1/6} P_2^{1/6} \left[P_1^{1/3} + P_2^{1/3}\right]^{\frac{-5}{2}}$
(D) none of these

10. The ratio of excitation potential of He^+ ion in first excited state to the ionization potential of Li^{2+} ion in its 3rd excited state is

(A) $\frac{16}{3}$
(B) $\frac{16}{9}$
(C) $\frac{8}{3}$
(D) $\frac{8}{9}$

11. The switch S is closed at t = 0. Now the switch is opened after the steady state is reached. Find the rate of change of current through the inductor just after the switch is opened. (there is no charge on the capacitor and no current in the inductor before closing the switch S).



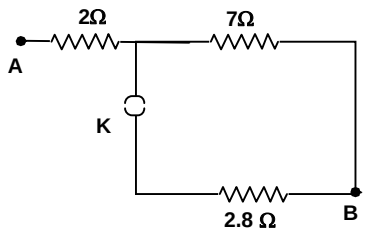
(A) $-\frac{V_0}{2L}$
(B) $\frac{V_0}{2L}$
(C) $-\frac{V_0}{L}$
(D) $\frac{V_0}{L}$

12. A metallic sphere is suspended in an evacuated chamber maintained at 200 K. The sphere is maintained at constant temperature of 600 K. A total of 280 W power is required to do it. When the sphere is blackened, 350 W power is needed to maintain the same temperature of the sphere. The emissivity of metallic sphere is
(A) 0.2
(B) 0.4
(C) 0.6
(D) 0.8

13. Water is pumped from a depth of 10 m and delivered through a pipe of cross section 10^{-2} m^2. If it is needed to deliver a volume of 10^{-1} m^3 per second, the power required will be ($g = 10$ m/s^2)
(A) 10 kW
(B) 9.8 kW
(C) 15 kW
(D) 4.9 kW

14. A bimetallic strip of total thickness x is straight at temperature θ. The coefficient of linear expansion of the two metals are α and 2α. Its radius of curvature when its temperature is raised to $\theta + \Delta\theta$ is
(A) $\frac{x}{4}\left[\frac{2+3\alpha\Delta\theta}{\alpha\Delta\theta}\right]$
(B) $\frac{x}{4}\left[\frac{2-3\alpha\Delta\theta}{\alpha\Delta\theta}\right]$
(C) $\frac{x}{4}\left[\frac{3+2\alpha\Delta\theta}{\alpha\Delta\theta}\right]$
(D) $\frac{x}{4}\left[\frac{3+2\Delta\theta}{\alpha\Delta\theta}\right]$

15. Power loss in the circuit shown is 1 watt (either key is closed or open) when a cell is connected between the points A and B. Then the emf of the cell in volt is
(A) 4
(B) 5
(C) 3
(D) 2



16. Two moles of an ideal diatomic gas follows 2dU = dW. Then the heat capacity of the gas is (The terms have their usual meanings)
(A) 7.5 R
(B) 15 R
(C) 4.5 R
(D) 9 R

17. A solid conducting sphere initially has a charge Q_0 on it. Now, the sphere is placed in an infinite medium having uniform conductivity σ and permittivity ε at t = 0. Then the charge on the sphere as a function of time t is
(A) Q_0
(B) zero
(C) $Q_0 e^{-\frac{\sigma t}{2\varepsilon}}$
(D) $Q_0 e^{-\frac{\sigma t}{\varepsilon}}$

18. A change of 8 mA in emitter current brings a change of 7.9 mA in the collector current. Find the values of β and α respectively.
(A) 79, 0.99
(B) 80, 0.99
(C) 80, 0.98
(D) 79, 0.98

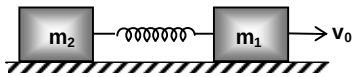
19. Apparent dip for the geographical meridian is 30° and the apparent dip for a plane which makes an angle with the magnetic meridian which is two times the angle between the geographical meridian and magnetic meridian is 45°. The declination is
(A) 15°
(B) 30°
(C) 45°
(D) 60°

20. The plano-convex lens is slivered from plane side and a point source of light is placed on the other side on its principal axis. Aperture diameter and maximum thickness of the lens are D and t respectively. (μ = refractive index of the lens). Then the distance of object from the silvered plane surface so that final image coincides with the object is (Assume t << D)
(A) $\frac{D^2 - 8(\mu-1)t^2}{8t(\mu-1)}$
(B) $\frac{D^2 - 4(\mu-1)t^2}{4t(\mu-1)}$
(C) $\frac{D^2 + 8(\mu-1)t^2}{8t(\mu-1)}$
(D) $\frac{D^2 + 4(\mu-1)t^2}{4t(\mu-1)}$

SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

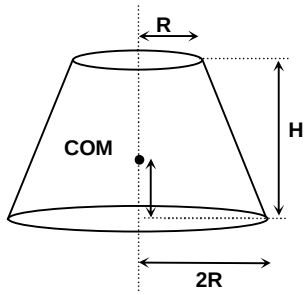
21. Two blocks A and B of masses 6 kg and 12 kg are connected with an ideal spring (k = 192 N/m) and placed at rest on the rough horizontal surface having coefficient of friction 0.2 for both the blocks. The minimum velocity given to the lighter block along the line joining them so that heavier block starts slipping in m/s is (g = 10 m/s^2) (spring is unstretched initially)



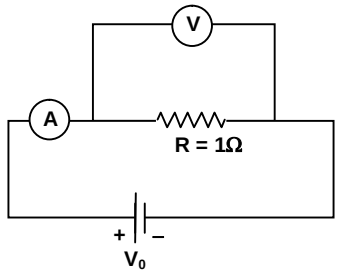
22. Two successive resonance are observed at length 53.8 cm and 75.8 cm in a resonance column tube. Then the radius of the tube is (in cm)

23. A uniform disc (R = 5m) is rolling without slipping on a horizontal surface such that a_{cm} = 5 m/s^2. The acceleration of point P which is at a distance of 4m from the centre of disc is zero. The angular velocity of the disc is $\sqrt{k}$ rad/sec. Find k.

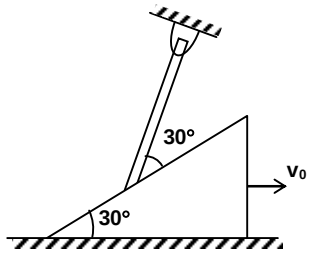
24. Centre of mass of uniform solid frustum (as shown) from its base is at a distance of h. If height of the frustum is H, then find $\frac{14h}{H}$.



25. Resistance of 1 Ω resistor is measured using Ohm's Law with the help of an ammeter and a voltmeter having resistance of 0.1 Ω and 100 Ω respectively as shown in the figure. Find the % error in measurement of resistance of the resistor is

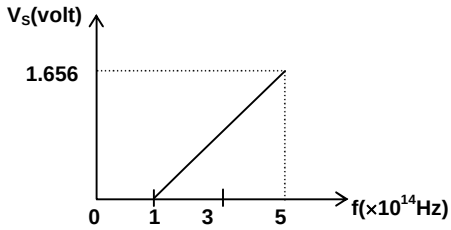


26. One end of a rotating rod slides on the moving wedge (as shown). The wedge is moving on the horizontal surface with a velocity v_0. If the angular velocity of the rod (length = L) is ω then the value of $\left(\frac{v_0}{\omega L}\right)^2$ at the instant shown is

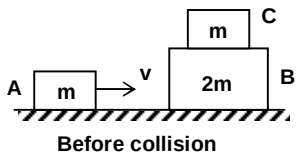


27. A boy runs away from a pole in a straight line so that its velocity is inversely proportional to the distance from the pole. When the boy is at a point A at a distance x_1 = 100 m from the pole, its velocity is v_1 = 2 m/s. What time (in seconds) will the boy take to run away from point A to point B which is at a distance x_2 = 200 m from the pole.

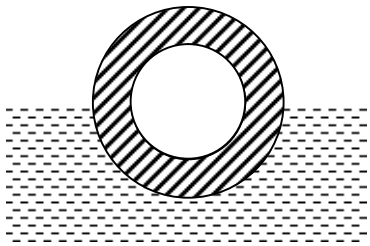
28. Figure shows the plot of the stopping potential (V_s) versus frequency f of the light used in an experiment on photoelectric effect. The ratio $\frac{h}{e}$ is m× 10^{-15} V-s. Then m is



29. A block A of mass m is projected with a velocity v as shown. The ground is smooth but there is friction between B and C. The resulting collision is elastic. Assume that C does not fall off B. If the final common velocity of B and C is $\frac{v}{n}$. Then find the value of n.



30. A thick shell of inner radius R_1 and outer radius R_2 floats with half submerged in a liquid as shown in the figure. If the density of the solid is two times the density of liquid then $\left(\frac{R_1}{R_2}\right)^3$ is



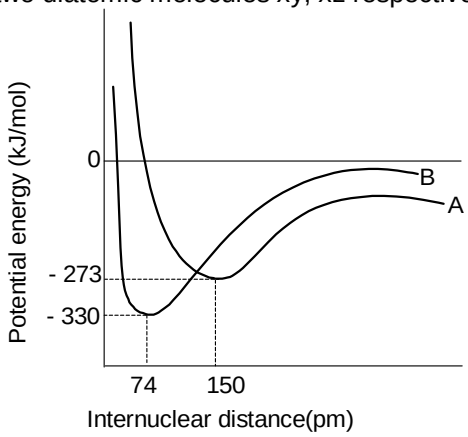
Chemistry

PART – B

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

31. The following diagram, potential energy vs. internuclear distance has been plotted for bonds of two diatomic molecules xy, xz respectively



The curves A and B respectively correspond to xy and xz. The correct statement is?

(A) Bond length of xy is lower than xz.
(B) Difference in bond lengths of xy and xz is 76 pm.
(C) Electronegative difference in xy is higher than that in xz.
(D) Both molecules have equal bond strength.

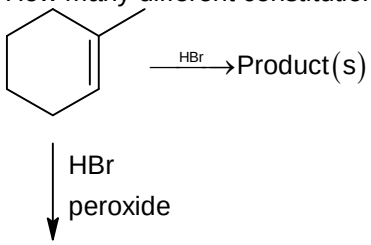
32. Which of the following statements about the given reaction is correct?

I. $I^- + CH_3 - x \longrightarrow CH_3 - I + x^-$

II. $F^- + CH_3 - x \longrightarrow CH_3 - I + x^-$

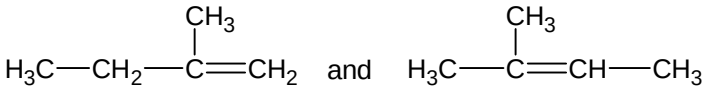
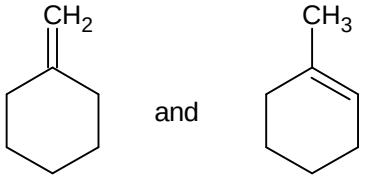
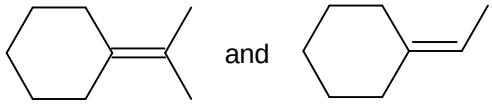
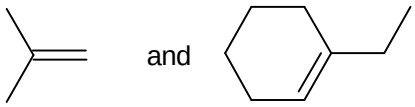
(A) The larger anions in protic solvents have poor nucleophilicity than smaller anions.
(B) Iodide ion can overlap with orbital of carbon from farther away, results in greater degree of bonding in the transition state.
(C) Fluoride ion can overlap with orbital of carbon from farther away, results in greater degree of bonding in the transition state.
(D) The nucleophilic character does not depend on solvent.

33. How many different constitutional isomers will be formed with the two reaction pathway?



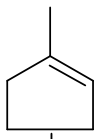
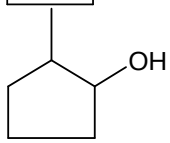
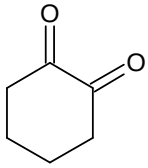
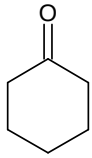
(A) 3
(B) 5
(C) 4
(D) 2

34. The complex formed by dissolving diborane in THF, called borane – THF complex, contains how many lone pair(s)?
(A) 2
(B) 1
(C) 0
(D) 3

35. In which of the following pairs of alkene, addition of HBr to second alkene is more highly regioselective?
(A) 
(B) 
(C) 
(D) 

36. $C_2H_5OCO(CH_2)_5CO_2C_2H_5 \xrightarrow[C_2H_5OH]{C_2H_5O^-} (A) \xrightarrow[2.\ \Delta]{1.\ H_3O^+} (B)$

$(B) \xrightarrow[H_2O,\ H^+]{CH_3MgBr} (C) \xrightarrow[\Delta]{conc.\ H_2SO_4} (D)$

(D) has degree of unsaturation equal to 2. Which of the following correctly matched with the product?
(A)
(D) = 
(B)
(C) = 
(C)
(B) = 
(D)
(B) = 

37. $CH_3-CH_2-\overset{\overset{18}{O}}{\overset{\|}{C}}-O-C_2H_5 \overset{H_3O^+}{\rightleftarrows} \underset{(Acid)}{(A)} + \underset{(Alcohol)}{B}$

$\left[\overset{18}{O} \rightarrow \text{isotopic oxygen}\right]$

For the de-esterification reaction, which of the following option is incorrectly matched?

(A) $A = CH_3-CH_2-\overset{O}{\overset{\|}{C}}-O-H$

(B) $A = CH_3-CH_2-\overset{O}{\overset{\|}{C}}-\overset{18}{O}-H$

(C) $B = CH_3-CH_2-O-H$

(D) $A = CH_3-CH_2-\overset{\overset{18}{O}}{\overset{\|}{C}}-\underset{18}{O}-H$

38. Two closed cubical vessels, one of them has twice the volume of the other, containing an ideal gas initially at pressure P_1 and temperature T_1 are connected through a narrow tube of negligible volume. The temperature of bigger vessel is increased to T_2. The final pressure P_f is:

(A) $P_f = 2P_i\left(\frac{T_2}{2T_1+T_2}\right)$

(B) $P_f = 3P_i\left(\frac{T_1}{T_1+2T_2}\right)$

(C) $P_f = 3P_i\left(\frac{T_2}{T_2+2T_1}\right)$

(D) $P_f = 2P_i\left(\frac{T_2}{T_1+2T_2}\right)$

39. Which of the following compounds will show increase in acidity constant when the solvent, in which it dissociates, is changed from water to a solvent which is less polar than water?

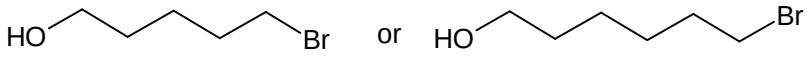
(A) CH_3CH_2COOH

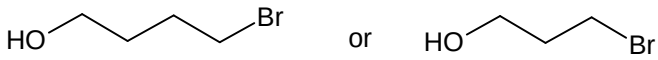
(B) $CH_3CH_2CH_2-OH$

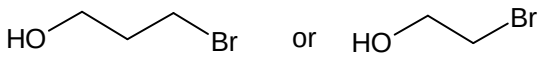
(C) Phenol

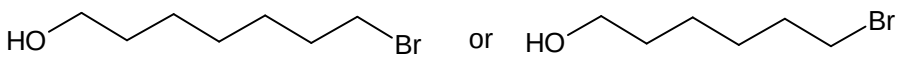
(D) $\overset{+}{R}NH_3$

40. After a proton is removed from the hydroxy group, in which of the following options, the second compound forms a cyclic ether more rapidly?

(A) 

(B) 

(C) 

(D) 

41. To attach methyl group with a nucleophile, methyl iodide would most likely to be used as the methylating agent. The correct reason can be
(A) iodide ion is a better leaving group.
(B) methyl iodide is liquid at room temperature while methyl chloride and methyl bromide are gases.
(C) iodide ion is the weakest base of the halide ions.
(D) all of the above.

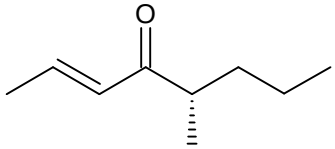
42. Which of the following reactions will better yield primary amine?
I. $CH_3 - CH_2 - CH_2 - Br \xrightarrow{NH_3}$
II. $CH_3 - CH_2 - CH_2 - Br \xrightarrow[H_2, Pd/C]{NaN_3}$
III. $NH_3 + CH_3 - I \longrightarrow$
(A) I only
(B) II only
(C) both I and III result primary amine equivalently
(D) both I and II result primary amine equivalently

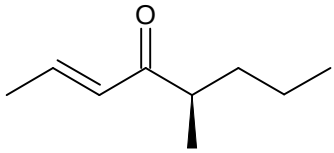
43. Find the n-factor of Fe_2S_3 in the following reaction
$Fe_2S_3 + 5O_2 \longrightarrow 2FeSO_4 + SO_2$
(A) 16
(B) 30
(C) 23
(D) 20

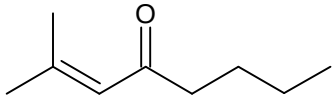
44. Choose the correct statements among the following
I. Excess amount of electrolytes in colloidal solution are preferred for the stability of the colloidal solutions.
II. Electrolytic impurities which are present in colloidal solutions are mostly in dissociated forms.
III. A positively charged sol is formed when ferric chloride is added to NaOH with adsorption of Fe^{3+} ions.
IV. Emulsion can not be diluted with any amount of the dispersion medium.
V. Among NaCl, $CaCl_2$ and $AlCl_3$; the minimum amount of electrolyte required to coagulate a certain amount of gold sol is in the case $AlCl_3$.
(A) I, II and V
(B) II and V
(C) II, III and V
(D) I, II and IV

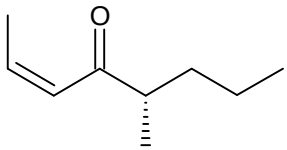
45. Order of bond lengths of the different bonds present in phenol is correctly shown in
(A) O – H > C – H > C – O > C - C
(B) C – C > C – O > C – H > O – H
(C) C – O > C – C > O – H > C - H
(D) C – O > O – H > C – C > C – H

46. (2E, 5R) – 5-methyl oct-2-en-4-one is

(A) 

(B) 

(C) 

(D) 

47. Which of the following can not give Fehling test?
(A) Formic acid
(B) 1-hydroxy-3-pentanone
(C) 1-Methoxy ethanol
(D) Acetaldehyde

48. If the temperature decreases below the freezing point of an aqueous solution then
(A) only solute molecules solidify
(B) only solvent molecules solidify
(C) vapour pressure of the solution is more than that was initially at the freezing point
(D) vapour pressure of the solution is more than that of pure solvent

49. Some oxides of p-block elements show colour while some don't, which of the following pairs of oxides has both oxides coloured?
(A) N_2O_4 and NO_2
(B) NO and N_2O
(C) NO_2 and N_2O_3
(D) NO_2 and N_2O_5

50. The separation of SO_2 and CO_2 can be done by using the reagent (s)
(A) lime water
(B) $BaCl_2$ solution
(C) $H_2O_2 + BaCl_2$ solution
(D) none of these

SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

51. Two Faraday of electricity is passed through a solution of $AlCl_3$. The aluminium deposited at the cathode is utilized in a separate experiment where it is oxidized to give aluminium oxide. The amount of aluminium oxide is recorded to be 8.5 g. Assuming the percentage yield during electrolysis was 50%. Find the percentage yield during this experiment?

52. The osmotic pressure associated with the fluid inside the blood cell is equivalent to that of 0.9% (mass/volume) sodium chloride solution. Suppose that the blood cell can withstand an extra pressure of 0.4205 atm along its cell wall. Find the maximum concentration (in % mass/volume) of an aqueous NaCl solution in which the blood cell should be put in without being ruptured. The value should be to the nearest integer. $\left(\text{Given}: T = 300\ K; R = 0.082\ \text{atm lit mol}^{-1}\ k^{-1}\right)$

53. The overall complex dissociation constant for the $\left[Zn(NH_3)_4\right]^{2+}$ is $x \times 10^{-12}$. Suppose the overall stability constant for this complex is 0.4×10^{11}.

54. 2 moles of an ideal gas at the state X (300 K, 2 atm) is subjected to the following change $(\text{assume } C_V = 1.5R)$

$$X \xrightarrow[\text{heating}]{\text{isochoric}} Y(5.0\ \text{atm}) \xrightarrow[\text{compression}]{\text{isothermal rev.}} Z(2.46\ \text{litre})$$

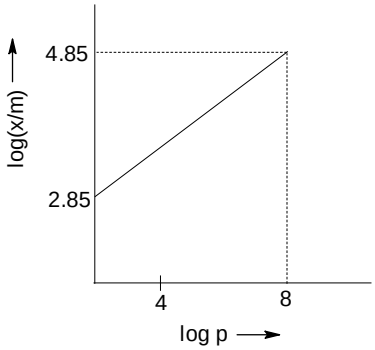
Find the magnitude of total work done during the complete processes in kJ.
$\left[\text{Given}: R = 0.082\ \text{atm Lt mol}^{-1}\ k^{-1}\right]$

55. 138 g of $N_2O_4(g)$ is placed in 8.2 Lt container at 300 K. The equilibrium vapour density of mixture was found to be 30.67. The degree of dissociation of $N_2O_4(g)$ is

$\text{Given}: \left(R = 0.082\ \text{atm lit mol}^{-1}\ k^{-1}\right); \left(N_2O_4 \rightleftharpoons 2NO_2\right)$

56. A photon whose frequency is equal to the frequency of third spectral line from red end of Balmer series in hydrogen spectrum. The photon is then absorbed by atom during excitation and the same atom releases two photons during de-excitation. One of the released photon has wave number equal to that of first line of Brackett series in hydrogen spectrum. Find the waver number (cm^{-1}) of the second photon. $\left(\text{Given}: R_\infty = \text{Rydberg constant} = 109677\ cm^{-1}\right)$

57. The following Freundlich isotherm has been plotted for the adsorption of a gas on some solid adsorbent at 300 K.



Find the mass of gas (in gm) which get adsorbed on 0.002 gm of the solid adsorbent at a pressure of 4 atm.

58. A closed vessel with volume 16.4 litre contains 100 g of $CaCO_3$. The vessel is then heated and $CaCO_3$ starts dissociating until the temperature reaches 1600 K and equilibrium is established. At equilibrium, 50% of calcium carbonate has bee dissociated. Find the dissociation constant $CaCO_3$ at 1600 K. $\left[CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)\right]$

59. A crystalline solid with formula unit AB shows NaCl type structure. Atom present at the body centre of the unit cell is replaced by another atom C. assuming charge neutrality maintained, find the mass of each unit cell in amu.
$\left[\text{Given}: M_A = 24 \text{ g mol}^{-1}, M_B = 32 \text{ g mol}^{-1} \text{ and } M_C = 9 \text{ g mol}^{-1},\right]$

60. At $27^\circ C$, the inversion of sucrose proceeds with a constant half-life of 500 min at pH = 5 for any concentration of sugar and 50 mins at pH = 4. The rate law for inversion of sucrose is given as
$\text{Rate} = k[\text{Sucrose}]^a \left[H^+\right]^b$
Find the value of b?

Mathematics

PART – C

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. Suppose $\log_x(2x^2 + x - 1) > \log_x 2 - 1$. Then the range of x is

(A) $\frac{1}{2} < x < 1$

(B) $x > \frac{1}{2}$ and $x \neq 1$

(C) $x > 1$

(D) $0 < x < 1$

62. Suppose $A = \{x \mid 5x - a \leq 0\}$, $B = \{x \mid 6x - b > 0\}$, $a, b \in N$, and $A \cap B \cap N = \{2, 3, 4\}$. The number of such pairs (a, b) is

(A) 20

(B) 25

(C) 30

(D) 42

63. Nine balls of the same size and colour, numbered 1, 2, ..., 9, were put into a packet. Now A draws a ball from the packet, noted that it is of number a, and puts back it. Then B also draws a ball from the packet and noted that it is of number b. Then the probability for the inequality $a - 2b + 10 > 0$ to hold is

(A) $\frac{52}{81}$

(B) $\frac{59}{81}$

(C) $\frac{60}{81}$

(D) $\frac{61}{81}$

64. Suppose real number a satisfies $|2x - a| + |3x - 2a| \geq a^2$ for any $x \in R$. Then a lies exactly in

(A) $\left[-\frac{1}{3}, \frac{1}{3}\right]$

(B) $\left[-\frac{1}{2}, \frac{1}{2}\right]$

(C) $\left[-\frac{1}{4}, \frac{1}{3}\right]$

(D) [–3, 3]

65. The absolute value of sum of all possible values of α for which the equation $|2x + 2|x| + 2\alpha| + |2x - 2|x| - 2\alpha| = 2016$ has exactly three solutions is

(A) 277

(B) 0

(C) 504

(D) 1008

66. Let f : R $\to$ [0, ∞] defined by $f(x) = \log\left(\sqrt{9x^2 - 12x + \lambda} + 1\right)$ be a onto function, where λ is a parameter, then complete set of values of λ is
(A) $(-\infty, 4]$
(B) $(4, \infty]$
(C) $\{4\}$
(D) $(-\infty, \infty)$

67. If $\int \frac{dx}{x\sqrt{1-x^{2021}}} = k \cdot \log\left|\frac{\sqrt{1-x^{2021}}-1}{\sqrt{1-x^{2021}}+1}\right| + c$, then the value of $\left(\frac{1}{k}\right)$ is equal to
(A) 2018
(B) 2019
(C) 2020
(D) 2021

68. If the value of $\lim\limits_{x\to 0^+}\left(\frac{(3/x)+1}{(3/x)-1}\right)^{1/x}$ can be expressed in the form of $e^{p/q}$, where p and q are relatively prime, then the value of (p + q), is
(A) 2
(B) 3
(C) 4
(D) 5

69. The range of $f(x) = \frac{\sqrt{x^2+1}}{x-1}$ is
(A) $\left(-\infty, -\frac{1}{\sqrt{2}}\right] \cup \left[\frac{1}{\sqrt{2}}, \infty\right)$
(B) $\left(-\infty, -\frac{1}{\sqrt{2}}\right] \cup \left(\frac{1}{\sqrt{2}}, \infty\right)$
(C) $\left(-\infty, -\frac{1}{\sqrt{2}}\right) \cup [1, \infty)$
(D) $\left(-\infty, -\frac{1}{\sqrt{2}}\right] \cup (1, \infty)$

70. Let $f(x) = \frac{x^2+2}{[x]}, 1 \le x \le 3$, (where [.] denotes greatest integer function), then incorrect statement is
(A) f(x) is increasing in [1, 3]
(B) least value of f(x) is 3
(C) f(x) has no greatest value
(D) f(2) = 3

71. A non-zero vector $\bar{a}$ is parallel to the line of intersection of the plane determined by the vectors $\hat{i}$, $\hat{i}+\hat{j}$ and the plane determined by the vectors $\hat{i}-\hat{j}$ and $\hat{i}+\hat{k}$. Then which of the following statement is correct?
(A) $\bar{a} = \frac{\hat{i}+\hat{j}}{\sqrt{2}}$

(B) $\frac{\hat{i}-2\hat{j}-2\hat{k}}{3}$

(C) Angle between $\bar{a}$ and $\hat{i}-7\hat{j}+2\hat{k}$ is $\frac{\pi}{2}$

(D) Angle between $\bar{a}$ and $\hat{i}-2\hat{j}+2\hat{k}$ is $\frac{\pi}{4}$

72. In a sequence of independent trials, the probability of success in one trial is $\frac{1}{4}$. Then the probability that the second success takes place on or after the fourth trail

(A) $\frac{17}{32}$

(B) $\frac{27}{32}$

(C) $\frac{53}{62}$

(D) $\frac{47}{62}$

73. If λ is a root of the equation $x^2 + 3x + 1 = 0$, then $\tan^{-1}\lambda + \tan^{-1}\frac{1}{\lambda}$ is equal to

(A) $\frac{\pi}{2}$

(B) $\frac{\pi}{3}$

(C) $-\frac{\pi}{2}$

(D) none of these

74. Let a^2, $a(b-3)$, $4b$ be the first three terms of on increasing AP, where $a, b \in I$ and $5 \le b \le 15$. Then the value of $\frac{b}{a}$ is equal to

(A) 1

(B) $\frac{5}{2}$

(C) $\frac{3}{2}$

(D) 7

75. The equation $|2z + 3| = |3z + 4|$ on the complex plane (where $z = x + iy$ and $x, y \in R$)

(A) represents a circle with centre $\left(\frac{6}{5}, 0\right)$ and radius $\frac{2}{5}$

(B) represents a circle with centre $\left(-\frac{6}{5}, 0\right)$ and radius $\frac{2}{5}$

(C) represents a circle with centre $\left(\frac{6}{5}, 0\right)$ and radius $\frac{1}{5}$

(D) represents a circle with centre $\left(-\frac{6}{5}, 0\right)$ and radius $\frac{1}{5}$

76. A sample of 35 observations has the mean 80 and standard deviation as 4. A second sample of 65 observations from the same population has mean 60 and standard deviation 3. The standard deviation of the combined sample is
(A) 5.85
(B) 5.58
(C) 10.12
(D) none of these

77. Let R and S be two equivalence relations on a set A. Then
(A) $R \cup S$ is an equivalence relation on A
(B) $R \cap S$ is an equivalence relation on A
(C) R – S is an equivalence relation on A
(D) none of these

78. Let p and q be two statements. Then, $(\sim p \vee q) \wedge (\sim p \wedge \sim q)$ is a
(A) tautology
(B) contradiction
(C) neither tautology nor contradiction
(D) either tautology or contradiction

79. Let the double ordinate PP′ of the hyperbola $\frac{x^2}{4}-\frac{y^2}{3}=1$ is produced both sides to meet asymptotes of hyperbola in Q and Q′. The product (PQ)(PQ′) is equal to
(A) 3
(B) 4
(C) 1
(D) 5

80. The director circle of the parabola $(y-2)^2 = 16(x+7)$ touches the circle $(x-1)^2 + (y+1)^2 = r^2$, then r is equal to
(A) 10
(B) 11
(C) 12
(D) none of these

SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

81. The two adjacent sides of a cyclic quadrilateral are 2 and 5 and the angle between them is 60°. If the area of the quadrilateral is $4\sqrt{3}$. Then the perimeter of the quadrilateral is

82. $\lim\limits_{x\to 0}\frac{1}{x}\int\limits_{-x}^{x}\left(t\cdot\sin\left(2010\,t\right)+2009\,t+1006\right)dt$ is equal to

83. The solution of the equation $x^2\,dy - y^2\,dx + xy^2\,(x-y)\,dy = 0$ is $\log\left|\frac{x-y}{xy}\right|=\frac{y^k}{2}+c$, then the value of k is

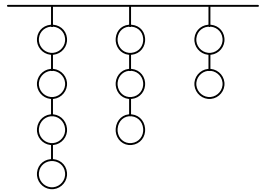
84. Let $M = \begin{bmatrix} a & -360 \\ b & c \end{bmatrix}$, where a, b and c are integers. Then the smallest positive value of b such that $M^2 = O$, where O denotes 2×2 null matrix is

85. The numbers of ways in which 6 balls can be selected from a bag containing 5 identical and 6 different balls is

86. If $(1+2x+3x^2)^{10} = a_0 + a_1x + a_2x^2 \ldots\ldots + a_{20} \cdot x^{20}$, then a_2 equals

87. A and B are two square matrices such that $A^2B = BA$ and if $(AB)^{11} = A^k \cdot B^{11}$. Then the value of k is

88. 9 clay targets are arranged as shown. Let N be the number of different ways they can be shot (one at a time) if no target can be shot until the target(s) below it have been shot. Then N is



89. If the locus of the point from where two of three normals to the curve $y^2 + 2y - 4x + 5 = 0$ are perpendicular is $\left(y - \frac{k}{2}\right)^2 - (x - \lambda) = 0$, then the value of $\lambda - 2k$

90. Let set X = {1, 2, 3, 4, 5} and set Y = {0, 1, 2, 3, 4, 5}, N be the total number of one-one function defined $f : X \to Y$ such that $f(1) \neq 0, 1$ and $f(i) \neq i$ for i = 2, 3, 4, 5, then N is